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A Comprehensive Visualization for Music Education and Artificial Intelligence

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***Abstract*—Artificial intelligence (AI) has revolutionized traditional methods and improved decision-making and automation. AI has also been used to enhance teaching methods, student learning, and research in music education. This study will examine literature on music education and AI. This study aims to investigate significant themes, trends, and achievements in this burgeoning discipline. This study will examine scholarly articles, conference papers, and other relevant literature to explore AI's applications, issues, and future in music education. Machine learning, natural language processing, computer vision, and deep learning are utilized in music education. These techniques are used in music composition, performance evaluation, instructional support, and individualized learning. Adaptive training, real-time feedback, and intelligent music production demonstrate the transformative potential of AI. This study will illuminate the obstacles AI faces in music education. Ethical considerations, data privacy, algorithmic bias, and human competence must be thoroughly investigated. In addition, the analysis would identify knowledge deficits for future research and development. This research could assist educators, researchers, and policymakers utilize AI in music education by conducting a comprehensive literature review. This work can assist in the development of AI-based instruments, the improvement of pedagogy, and the promotion of music education.**

***Keywords*—Artificial intelligence; bibliographic analysis; deep learning; machine learning; music education.**

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I. INTRODUCTION

The study and practice of teaching and learning music are part of the field of music education. It is essential for growing people's musical skills [1], encouraging creativity [2], and helping people understand other cultures [3]. In recent years, there has been a growing interest in how music education and artificial intelligence (AI) can work together to improve learning and teaching [4]–[7]. Bibliographic analysis is a way to do research that includes looking at and judging the existing literature on a particular subject [8]. Its goal is to give a complete picture of the state of knowledge, point out trends and research gaps, and show how a particular area has grown and changed [9]. Through bibliographic analysis, researchers can learn more about the existing body of work, assemble what they already know, and find possible directions for future study.

Artificial intelligence in music education has much potential to improve many different parts of the learning process [10]. AI technologies can make learning more personalized and adaptable, help with writing and analyzing music, make intelligent training systems possible, and open the door to new ways of teaching music. These technologies use machine learning algorithms [11]–[15], natural language processing [16], computer vision [17], Blended learning [18], and other AI methods to process and analyze musical data, make music, and give intelligent feedback.

A thorough look at the research on music education and AI shows that there are a few key trends and study themes. Researchers have looked into how AI could be used in music education, such as learning to play an instrument [19], [20], music theory, writing, improvisation, and group performance. Researchers have examined how well AI-based tools and apps help people learn music, become interested in it, and be creative. They have also examined how AI affects how music is taught, how teachers do their jobs, and how AI tools are used in classrooms.

Even though interest is growing and there may be benefits, using AI in music education is still a new and changing area. Some study gaps and problems need to be looked into more. These include questions about the availability and quality of data, the creation of easy-to-use AI tools for music education, and the ethical aspects of using AI [21], and the need for real-world studies to determine how well AI-based interventions help people learn music in the real world. Music teachers, AI researchers, cognitive scientists, and business people must work together to move the field forward and solve these problems.

This research aims to explore and discuss music education-based artificial intelligence-based bibliographic analysis to give valuable information about the field's current research, trends, and challenges. It would be beneficial to show how AI technologies have the potential to change music instruction and make learning more fun. However, more studies are needed to solve the existing problems and ensure that AI tools can be used effectively in music education. By filling these gaps, experts can help make AI-based solutions that help people of all ages and skill levels learn and teach music better.

A. Literature Review

The rapid development of technology, particularly artificial intelligence (AI), has sparked interest in the convergence of music education and AI [5]. This literature review examines artificial intelligence research in music education. AI can enhance music education's teaching and learning. Using machine learning, deep learning, and natural language processing, intelligent tutoring systems, virtual music mentors, and interactive learning platforms were developed [22]. These applications enhance music education through personalized training, real-time feedback, and adaptive learning environments [23]. AI-based systems can provide students with customized feedback, identify errors, and designate skill-specific practice tasks [10], [24], [25]. AI can aid in composing, improvising, and analyzing music, thereby expanding creative possibilities [23], [26].

However, a dearth of human interaction, excessive reliance on technology, and the need for trained instructors to implement AI tools must be acknowledged. These systems use AI to provide individualized instruction to students. These systems evaluate students' performance data, identify areas for advancement, and provide adaptive content and activities. Chatbots and virtual mentors can simulate human interactions [27], facilitating interactive learning environments.

AI algorithms contribute to innovative music composition. Artificial intelligence models can learn from music [20], analyze patterns [19], and create new melodies and harmonies [28]. Students can experiment with composition, musical genres, and originality using these systems. AI-based techniques have been implemented for music analysis and recognition. Transcription, genre classification, instrument recognition, and emotion detection of music are performed automatically. These apps aid music teachers in analyzing pupil performances, gauging their development, and providing constructive feedback. Approaches based on constructivism emphasize active participation, collaborative learning, and inquiry-based learning. AI technology can scaffold, guide, and support students' musical discovery in constructivist frameworks. AI can facilitate collaboration among music students. Students can communicate, share compositions, provide peer feedback, and create music on AI-enabled platforms. These platforms facilitate community, collaboration, and education.

Artificial intelligence in music education has tremendous potential, but numerous problems must be resolved. These objectives include creating ethical rules for the use of AI, evaluating and enhancing AI systems, educating educators on how to use AI tools, and assessing AI's social influence on music education. This review examines the literature on music education and artificial intelligence. The application of AI in music education could enhance teaching and learning. Intelligent teaching systems, music composition generators, and music analysis algorithms provide individualized instruction, creative exploration, and real-time feedback. Adoption of success requires addressing human contact, excessive technological use, and the need for trained educators.

II. MATERIAL AND METHOD

Our research starts by searching the Scopus database for articles on music education and artificial intelligence. The analysis was based on and modified by Donthu et al. [9], as seen in Figure 1.

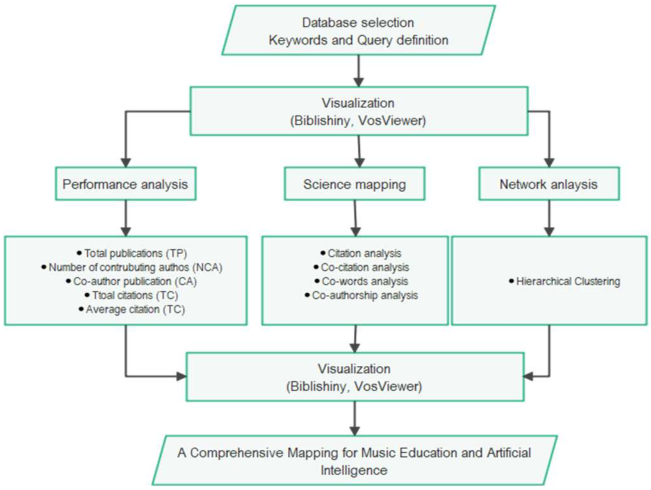

Fig. 1 Research step and design

In the first stage, we create a search operator and keyword strategy with keywords "Music education" AND ("artificial intelligence" OR "Deep learning" OR "machine learning") by combining with the use of Boolean operators (AND, OR) to refine the search. Then, we standardize and correct bibliographic information by removing duplicates, standardizing author, title, and journal names, and correcting data errors. We will use the Biblioshiny library on Bibliometrix [29] and VosViewer [30], [31] to analyze and visualize the selected data sources. Biblioshiny is an R Shiny application for interactive bibliographic data analysis and visualization. VOS viewer imports clean and standardized bibliographic data. Data can be read from a CSV. The results would be the most cited papers, the top authors, and key research issues, using network visualization, term clustering, co-occurrence patterns, and citation relationships to comprehend music education and AI research.

III. Results and Discussion

B. Performance Analysis

1) Data sources overview, documents, and authors: We used the query of (TITLE-ABS-KEY("Music education" AND ("artificial intelligence" OR "Deep learning" OR "machine learning"))) on the Scopus database to find articles related to artificial intelligence and music education topics. The results were 162 documents published from 1986 to 2024 (Table 1). The category was 88 articles, and 61 conference papers exemplify ongoing research and novel ideas in the topic, as evidenced by professional debates and presentations. The six-book chapters demonstrate extensive research on this issue in detailed writing. Three retracted articles demonstrate the dynamic and self-correcting nature of scholarly discourse. One book in the list offers a thorough and all-encompassing summary. Conference evaluations, errata, and reviews provide unique perspectives and updates to intellectual discourse.

TABLE I
Data sources, document, and authors information

Publication Timespan (1986:2024)	Document Contents	Authors	Authors Collaboration
Sources (Journals, etc) = 90 Documents=162 Annual Growth Rate %=0 Document Average Age=2.94 Average citations per doc= 3.296 References= 3870	Keywords Plus (ID)=952 Author's Keywords (DE)=390	Authors = 297 Authors of single-authored docs=68	Single-authored docs=73 Co-Authors per Doc=2.08 International co-authorships %= 9.877

A total of 162 documents were published in 90 journals and books within the time stamp. Although there is a significant quantity of publications, the annual growth rate remains at 0%, indicating a consistent rate of publication. The materials, with an age of 2.94 years, are quite recent, suggesting ongoing research and importance. The average number of citations per paper is 3.296, suggesting a moderate level of academic impact and engagement. A multitude of sources (3870) demonstrate extensive investigation and citation. The 952 Keywords demonstrate the research's extensive and specialized focus areas Plus (ID) and 390 Author's Keywords (DE) found in the documents. Out of the total 297 contributors, 68 have produced documents as sole authors, indicating a combination of independent and collaborative research. Out of the papers, 73 were written by a single author, while the average number of co-authors per document is 2.08, suggesting a significant amount of collaborative research in this area. International co-authorships are few, accounting for only 9.877% of documents. This data provides insights into the patterns of publication, research areas of interest, and collaborative tendencies within this particular discipline.

2) The trend of publication between 1986 and 2024: Figure 1 shows the results of the publication between 1986 and 2024, demonstrating a captivating progression in both the quantity of published publications and their subsequent influence in terms of citations. From 1986 to 2018, no citations were recorded for any documents, indicating that these works, although they existed, did not have a substantial impact or involvement with the academic community. However, there was a significant change in this pattern in 2019. The number of published publications climbed to 7 and received 20 citations, suggesting a growing acknowledgment and importance in the field. The trend of increasing citations continues, as evidenced by the number of papers and citations received in recent years. In 2020, 17 documents received a total of 33 citations. This trend further intensifies in 2021 and 2022, with 21 and 47 documents, respectively, earning 55 and 129 citations. In 2023, there is a noticeable peak as 41 documents gather a remarkable 190 citations, indicating significant involvement and impact within the academic community. Notably, there is a data point for the year 2024, which suggests the presence of a single article that has not received any citations. This indicates that the text may have been published relatively recently and has not had the time to be referenced by other works. In summary, this data indicates a rising interest and significance in this specific area of research, characterized by a gradual rise in both the number of publications and their academic influence over the years.

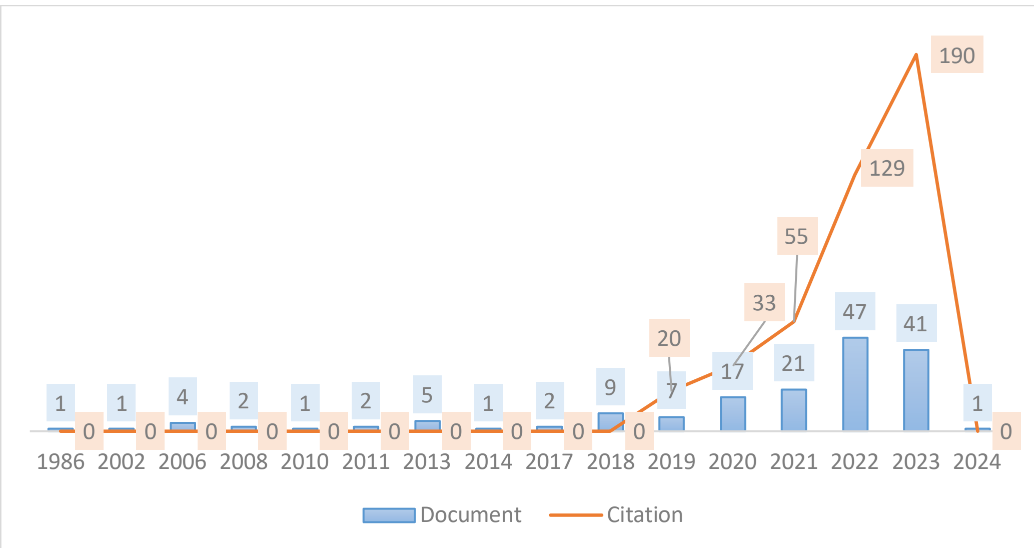

Fig. 2 Data source results

C. Science Mapping

1) Citation and Authorship Analysis: The citation data provides information on research works' academic impact and temporal significance (Table 1). The initial work by Deliège I, which was published in 2006, has garnered a total of 60 citations, with an average of 3.33 citations each year. The normalized overall citation count of 3.04 indicates enduring recognition throughout time. Pati Ka's 2018 article demonstrates its ongoing importance with 24 citations and an average of four per year. Huang C-Za's work in 2019 has received twenty-two citations, with an average of 4.4 citations each year. The normalized citation count of 3.85 indicates a significant increase in influence. The academic importance of Dalmazzo D's work in 2017 and Holland S's publication in 2013 is evident through their respective normalized citation counts of 1.9 and 2.43.

TABLE II
MOST GLOBAL CITED DOCUMENTS

Reference	Total Citations	TC per Year	Normalized TC
[32]	60	3.33	3.04
[4]	29	29	29.73
[33]	24	4	4.24
[34]	22	4.4	3.85
[10]	20	10	8.7
[35]	20	2.86	1.9
[36]	18	1.64	2.43
[37]	15	1.36	2.03
[38]	13	3.25	3.45
[39]	12	4	3.65

Yu's [4] work in 2023 is notable for its rapid impact, as evidenced by its 29 citations in the same year and a high normalized citation count of 29.73. This indicates swift and substantial acknowledgment in the field. The 2022 publication by Wei [10] garnered 20 citations and achieved a high normalized count of 8.7 in its inaugural year. By comparison, the article by Abeßer J in 2013 and the study by D'amato [38] in 2020, they had a consistent and modest academic impact, as evidenced by their normalized citation counts of 2.03 and 3.45, respectively. Huang C's work in 2021 received 12 citations and has a normalized count of 3.65, suggesting a positive initial reception and potential relevance. These citation metrics offer valuable information on the significance and durability of scholarly works within their respective fields.

Figure 3 displays the fractionalized article counts and article counts of the writers. Wang X is at the top of the list with five publications, which may be divided into fractions of 3.7, indicating the presence of several authors. Dalmazzo [35] has authored three papers, with a fractionalized count of 1.33, indicating a slightly lower share of authorship than the total number of publications. A higher fractionalized count of 2.33 for LIU J's three articles suggests a more immense individual contribution. The three papers authored by Volpe G have a fractionalized count of 0.83, indicating a high level of collaboration. Other authors, such as Anguita D, Camurri A, and D'amato [38], have contributed to 2 papers and have a fractionalized count of 0.33, indicating a noteworthy level of co-authorship. Delgado [40], [41] has written two articles, each with a fractionalized count of 0.67. On the other hand, Burrows [42], [43] has also written two articles, each with a fractionalized count of 0.75. Chen [44], [45] has two publications with a fractionalized count of 1.2, suggesting a significant individual contribution. The distribution of publications and fractionalized counts illustrates the collaborative nature of academic publishing, where multiple authors frequently contribute to a single piece.

Figure 4 compares total citations (TC) and average article citations by country. China has the highest number of citations, totaling 216. However, the average citation per item is only 2.6, indicating many publications with a relatively low citation rate. Georgia has the most significant average citation rate at 13, indicating that although it has a lesser research output, each work has a considerable impact. A total of twenty-six citations support this. Italy ranks second with 23 citations and an average of 4.6 per article, demonstrating a balance between output and effect.

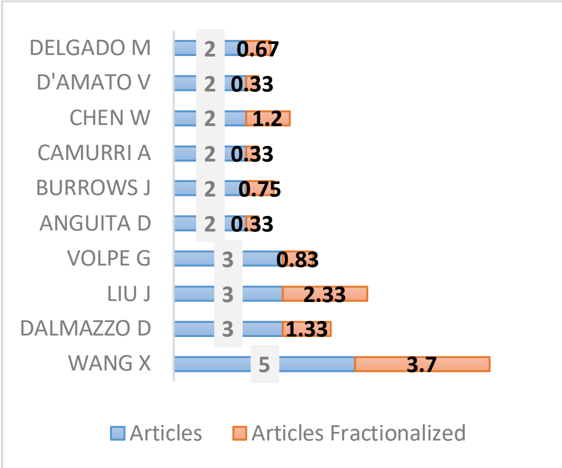

Fig. 3 Most Relevant Authors

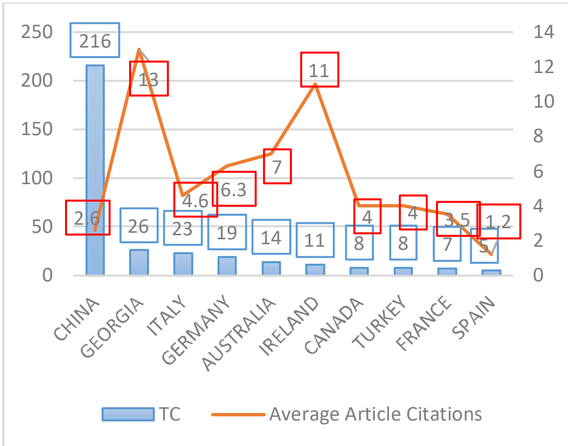

Fig. 4 Most Cited Countries

Germany and Australia had a total of 19 and 14 citations, respectively. However, their average article citations of 6.3 and 7 suggest that their research articles are mentioned more frequently. Ireland, with 11 citations, is comparable to Georgia in terms of average article citations, indicating its noteworthy research importance. Canada and Turkey have obtained eight citations, with an average of 4 citations per article, suggesting a moderate level of influence. France has a total of 7 citations, with an average of 3.5 citations for each piece. In contrast, Spain has the lowest average of 1.2 citations per item, despite having five citations. The data demonstrates variations in research impact and productivity across different countries. Some countries prioritize producing a large volume of research, while others focus on the academic influence of each paper.

Figure 5 shows clusters of nodes, which are most likely research papers or authors, as shown by a network analysis. Network centrality measures in the nodes are evaluated using the metrics of Betweenness, Closeness, and PageRank. Betweenness centrality quantifies how much a node lies on the shortest pathways between other nodes. Nodes with a value of 0 are not considered significant network bridges. Greater values of proximity centrality indicate reduced average distances from a node to all other nodes in the graph. Clusters 1 (red line) and 2 (blue line) have a proximity centrality of 0.2, indicating their core position in the network's topology. Conversely, cluster 4 (purple line) has a centrality of 0.1, and cluster 5 has a centrality of 0.02, suggesting their peripheral nature. Every node inside this network possesses an identical PageRank value of 0.05, signifying an equivalent level of importance or effect determined by the PageRank algorithm. Clusters 1 and 2, with their high closeness centrality, likely correspond to cohesive and key themes or issues within a more extensive academic network. On the other hand, clusters 4 and 5 (Orange line) may indicate more specialized or isolated research fields. This method facilitates the identification of the structure and interconnectedness of academic research.

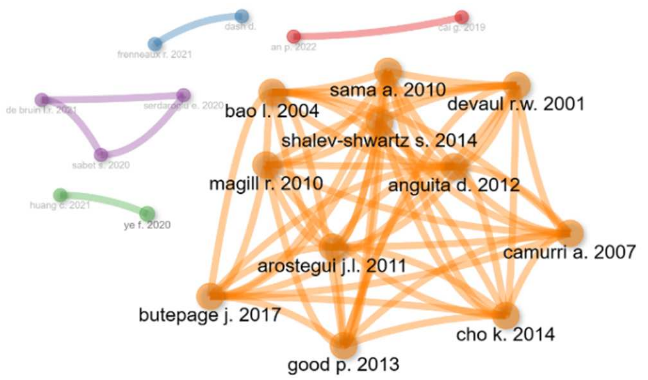

Fig. 5 Author co-citation network

2) Co-occurrence analysis based on keywords: Co-occurrence analysis is a method employed to examine the associations between subjects in a collection of literature, typically represented by keywords or index terms derived from a set of texts. Every row in this study corresponds to a phrase with attributes that measure its presence and impact in the dataset. Ultimately, this co-occurrence analysis demonstrates the interconnection between subjects, their frequency, recentness, and implications for research. It aids in identifying crucial and emerging areas of study and comprehending scholarly discussions.

The clusters identified by co-occurrence analysis represent areas of high thematic density within the research landscape. Each cluster comprises correlated terms commonly occurring in the same publications (Figure 6). Furthermore, the presence of distinct clusters can provide evidence for several subfields or specializations within the area, while highlighting its multidisciplinary nature.

Cluster 1 appears to represent a category centered around the practical applications of technology, as shown by terms such as application programs [46], computer-aided education [47], and e-learning [48]. Computer Music [26] and music information retrieval [49] are two examples that highlight a specific emphasis on music technology in educational environments (Figure 5). This cluster seems to be linked to educational technology and its application in learning environments. Cluster 2 focuses on the fundamental and theoretical components of technology, as indicated by the inclusion of terms like computer technology [50], education computing [51], and information management [52]. Considering the use of the terms human [53] and humans [54], it is plausible that a technology-focused approach is being adopted, which might entail studying the interaction between humans and technical systems. The terms 5G mobile communication systems [55] and big data [56] are prevalent in Cluster 3, which focuses on communication and data technology. The inclusion of the phrase education systems [57] and students [58] within this cluster suggests a potential research interest in the educational applications of communication technologies. Cluster 4 stands out for its use of advanced computational techniques, as seen by terms like audio acoustics [59], convolution [60], and deep learning [61]. These approaches are applied in several domains, including music and acoustics. The presence of both music [62] and musical instruments [63]–[65] suggests a strong association with music technology.

The weight of the links and the frequencies of terms inside clusters offer further context. A high frequency of links and occurrences inside a cluster indicates that a term holds a central position within the cluster, signifying a key focus of attention or a significant subject extensively researched and discussed. The average number of citations indicates the impact and acknowledgment of the cluster's subjects throughout the academic world. On the other hand, the average publication year within each cluster offers insight into the evolution of the cluster's themes over time. Each cluster is a research study group with standard themes and keywords. These clusters provide a clear overview of different study topics, their relative importance, recency, and impact in the broader academic discourse. Put, each cluster is a concise overview of the conducted research.

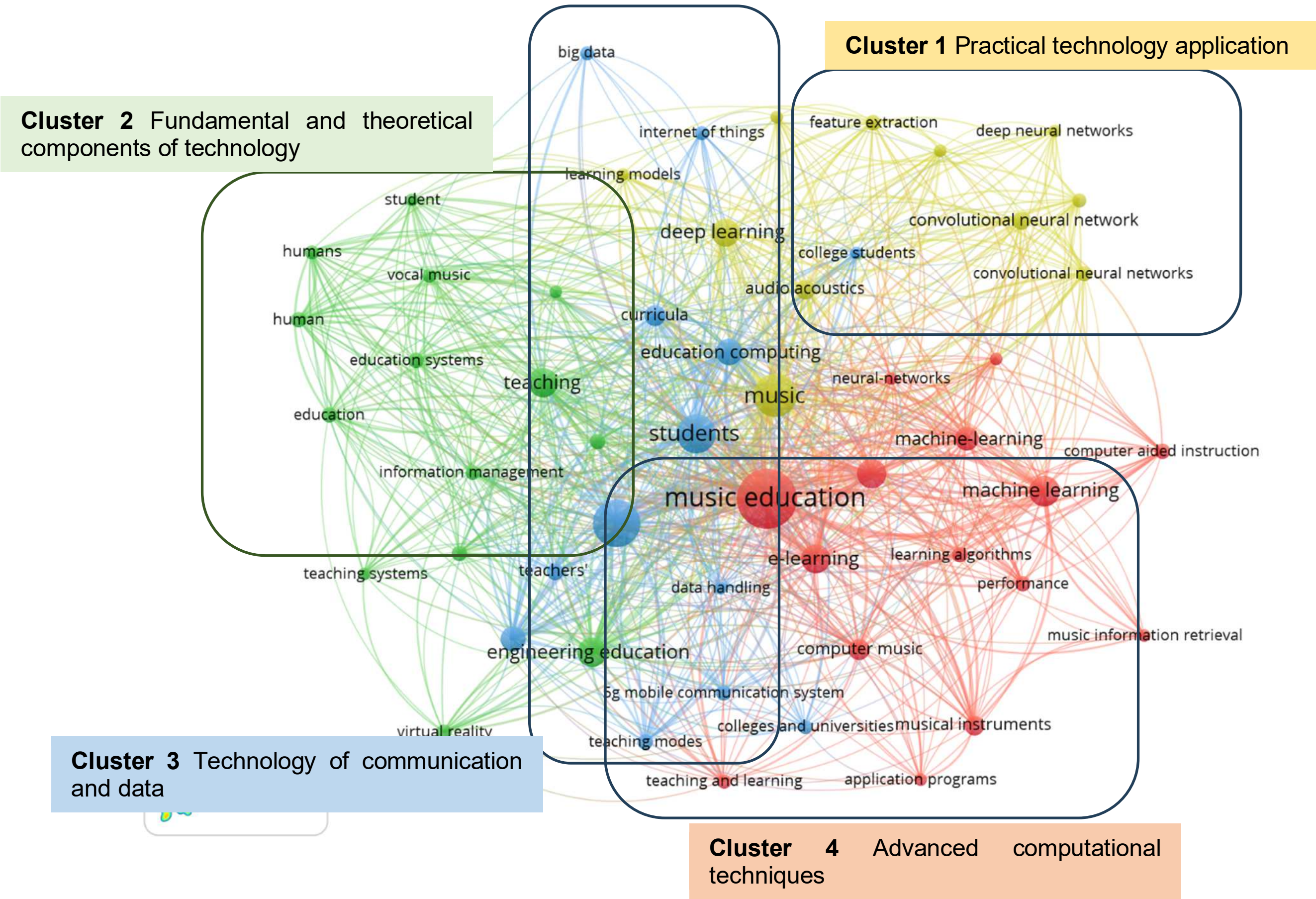

Fig. 6 Co-occurrence analysis

D. Network Analysis

Figure 7 comprehensively analyzes countries' research production and influence, including weighted criteria and evaluations. China possesses the highest link strength, which signifies substantial collaboration or citation connections, and the highest document weight, suggesting a significant amount of research output. The UK and Belgium have a smaller but highly referenced body of work with fewer links and documents. In contrast, countries such as Brazil and South Korea have lower Citations, which suggests a lower level of research impact. Other countries, such as the United States and France, maintain a balance between the number of documents and citations, resulting in average publication years and citations. This indicates a steady research output with a consistent level of influence. Spain signifies a substantial citation, suggesting that Spain has a higher impact than other published works on the same cluster. This analysis assesses the productivity and importance of research conducted by each country in the dataset, as well as their collaborative networks and relative contributions to the field.

Fig. 7 Countries' bibliographic coupling

Bibliographic coupling analysis offers valuable insights into the organization of scientific communication and the spread of knowledge within and between different research fields. This approach emphasizes the interdependence of scholarly journals and their respective impact within the academic sphere. Figure 8 represents a bibliographic coupling analysis, a technique used to examine the connections between journals by analyzing the frequency of citations between articles published in each journal and related journals. Acknowledging that the links represent the quantity of connections a journal has with other journals is crucial. Bibliometrically, this pertains to the number of references often found among different publications. For instance, the "Wireless Communications and Mobile Computing" field has the highest number of standard references, with a link strength of 6 and a total link strength of eleven. From this, it seems that this publication possesses a robust network of affiliations with other publications that pertain to its subject matter, suggesting its significance in the realm of wireless and mobile technology.

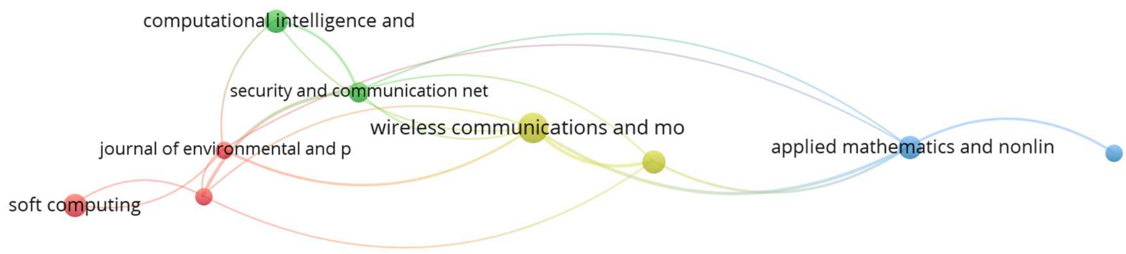

Fig. 8 Source bibliographic coupling

Journals of "Computational Intelligence and Neuroscience" and "Interactive Learning Environments" both weight 3 but differ in their total link strength. Although they have the same number of connecting journals, the depth or frequency of those connections may vary. Furthermore, these journals have a commendable level of citations relative to their recent average publishing years 2022, suggesting a growing impact within their respective domains. "Applied Mathematics and Nonlinear Sciences" lacks both citations and normalized citations. The journal may be in its initial phases, as its articles have not been referenced yet, or there is a deficiency in connecting with the existing literature on the topics it focuses on.

The lower link strength and citation metrics of specific journals, such as "Security and Communication Networks" and "Journal of Environmental and Public Health," may indicate that they are more specialized or emerging within their respective research fields, or that they focus on subjects that have less connection to the broader body of literature. Bibliographic coupling is an efficient method for evaluating scientific areas' conceptual framework and cohesiveness. Based on the average publication year and citation rates, it can be observed that highly connected journals tend to focus on comparable scopes or subjects. These publications can indicate established research fronts or emerging trends depending on the research findings. This viewpoint considers factors such as the journal's magnitude and citation procedures in various disciplines.

E. Thematic Trend

Further examination based on the thematic map indicates that continuous research is being conducted in these groups and suggests the emergence of new patterns and the intersection of multiple disciplines (Figure 9). The significant centrality and density of deep learning [66] indicate its growing impact and unified progress, potentially influencing intelligent computing [67]–[69] and virtual reality [70]–[72]. The high frequency, centrality, and density of the term 'music education' suggest that it is a significant and dynamic area of study, which could potentially influence related clusters such as 'musical instruments' and 'deaf music education'.

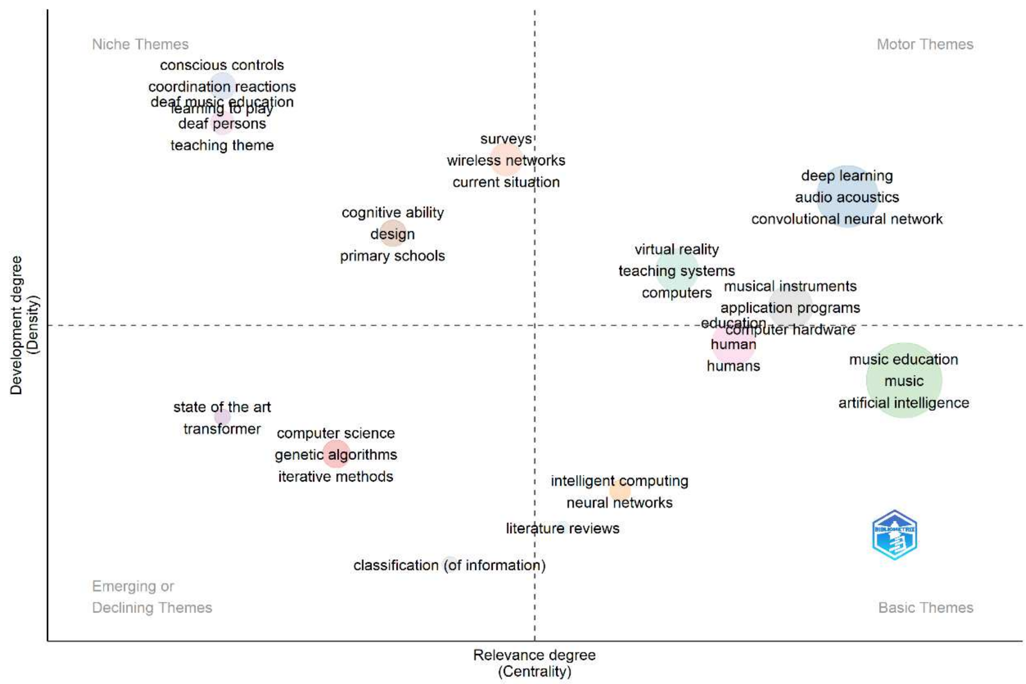

Fig. 9 Thematic map

Clusters such as state-of-the-art [73] and conscious controls [38], [69] exhibit a centrality of 0%, indicating that they are peripheral or nascent domains with few connections to the research network. Although surveys [74], [75] and conscious controls exhibit lower centrality, they possess a high density, indicating that while they may be internally cohesive, they have not yet established robust external research connections. The varying levels of centrality and density observed within clusters indicate the dynamic nature of academic research, where specific fields may exhibit strength but have limited connections to others, and vice versa. Understanding the emergence, interaction, and influence of many disciplines of study is crucial for comprehending the development of scientific knowledge. Additionally, it assists academics in identifying influential locations, collaborators, and emerging areas of study.

F. Limitations and Future Directions

Integrating artificial intelligence (AI) into music education has limitations that must be addressed to harness its potential fully. A significant limitation is the restricted availability and stringent data quality requirements for training artificial intelligence algorithms. High-quality, diverse datasets are crucial for the progress of AI solutions that are both effective and unbiased. There is a lack of comprehensive statistics covering the wide range of music instruction situations. Furthermore, there is a necessity for AI tools that possess user-friendly interfaces and are specifically tailored for music education. Many existing technologies may be excessively complex for educators lacking technological proficiency, limiting their accessibility and general adoption.

Ethical considerations, especially regarding data privacy and the potential reduction of human engagement in the learning process, are of paramount relevance. Given the growing ubiquity of AI tools, it is imperative to ensure that they augment rather than replace the fundamental human elements of teaching and creativity. Moreover, there is a dearth of empirical evidence regarding the efficacy of AI interventions in real educational settings. Further investigation is necessary to understand the proper incorporation of artificial intelligence (AI) in the classroom and evaluate its impact on music learning outcomes.

An interdisciplinary approach is crucial for identifying future research trajectories. The collaboration of music educators, AI developers, cognitive scientists, and industry practitioners can produce AI solutions that are both pedagogically impactful and technologically dependable. This partnership can also ensure that AI technologies are created with great regard for ethical implications. Moreover, there is a need for further empirical research to assess the effectiveness of AI-driven music education interventions in different real-world settings. The aim of this research should be to understand the contextual factors that influence the effectiveness of artificial intelligence in education, such as the learning environment, the learners' characteristics, and the educators' readiness to adopt new technology. To address these limitations and give priority to these potential areas of research, the field can make progress in artificial intelligence (AI) technologies that enhance music education for individuals across different age groups and skill levels.

IV. Conclusion

Our bibliographic analysis of the interface between music education and artificial intelligence (AI) has yielded valuable insights into the present research landscape, elucidating trends, problems, and new paths for progress. Our research indicates that the incorporation of artificial intelligence (AI) into music education is rapidly growing. There is evidence suggesting that AI has the potential to customize learning experiences and improve creative processes. However, it is essential to note that this sector is still in its early stages, and various obstacles must be addressed. The obstacles encompass the limited availability of data, the requirement for AI tools that are easy to use, and ethical considerations. Our analysis indicates that although AI presents intriguing technologies for music education, its implementation depends on addressing these limitations. Future research should prioritize interdisciplinary partnerships to create morally principled, empirically substantiated artificial intelligence (AI) instruments that can be seamlessly included in music education methodologies, thereby enhancing the educational process and nurturing the abilities of a wide range of learners.

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